

Statistical Models

Assignment 2

Thomas Sejer Canham

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Introduction

We are given a geometrically distributed random variable X with PMF:

$$P(X = k) = (1 - \pi)^{k-1}\pi, \quad k = 1, 2, 3, \dots, \quad 0 < \pi < 1 \quad (1)$$

And we wish to find the moment generating functions, its relation to the negative binomial distribution, and its limiting distribution as $\pi \rightarrow 0$.

a)

We wish to find the moment generating function of X . The MGF is defined as

$$M_X(t) = E[e^{tX}] = \sum_{k=1}^{\infty} e^{tk} P(X = k) = \sum_{k=1}^{\infty} e^{tk} (1 - \pi)^{k-1} \pi. \quad (2)$$

We can factor out π and rewrite the sum

$$M_X(t) = \pi \sum_{k=1}^{\infty} e^{tk} (1 - \pi)^{k-1} = \pi e^t \sum_{k=1}^{\infty} (e^t (1 - \pi))^{k-1}, \quad (3)$$

now let $q = (1 - \pi)$, then we have

$$M_X(t) = \pi e^t \sum_{k=1}^{\infty} (e^t q)^{k-1}. \quad (4)$$

We can factor out one division of q of the entire sum

$$M_X(t) = \frac{\pi}{q} \sum_{k=0}^{\infty} (e^t q)^k \quad (5)$$

Using the hint, we get a geometric series if $|e^t q| < 1$, since there is a product of strictly positive numbers, we can drop the absolute value, such that

$$e^t q < 1 \implies t + \ln(q) < 0 \implies t < -\ln(q) = -\ln(1 - \pi). \quad (6)$$

Using the formula for the sum of a geometric series, provided we satisfy (6), then

$$M_X(t) = \frac{\pi}{q} \cdot \frac{e^t q}{1 - e^t q} = \frac{\pi e^t}{1 - (1 - \pi)e^t} \quad (7)$$

We can now use this moment generating function to find the mean and variance of X

$$\frac{d}{dt}(M_X(t)) = \frac{\pi}{q} \frac{d}{dt} \left(\frac{e^t q}{1 - e^t q} \right) = \frac{\pi}{q} \cdot \frac{q e^t}{(1 - q e^t)^2} = \frac{\pi e^t}{(1 - q e^t)^2} \quad (8)$$

Setting $t = 0$, we get

$$\left. \frac{d}{dt}(M_X(t)) \right|_{t=0} = \frac{\pi}{(1 - q)^2} = \frac{\pi}{\pi^2} = \frac{1}{\pi} \quad (9)$$

We can then find the variance using the formula $\text{Var}(X) = E[X^2] - (E[X])^2$. Using that $E[X]^2 = \frac{1}{\pi^2}$. We can find $E[X^2]$ by taking the second derivative of the moment generating function

$$\frac{d^2}{dt^2}(M_X(t)) = \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) = \frac{d}{dt} \left(\frac{\pi e^t}{(1 - qe^t)^2} \right) = \frac{\pi e^t (qe^t + 1)}{(1 - qe^t)^3}. \quad (10)$$

If we set $t = 0$, we arrive at

$$\left. \frac{d^2}{dt^2}(M_X(t)) \right|_{t=0} = \frac{\pi(1+q)}{(1-q)^3} = \frac{\pi(q+1)}{\pi^3} = \frac{(q+1)}{\pi^2}. \quad (11)$$

as such

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{(q+1)}{\pi^2} - \frac{1}{\pi^2} = \frac{q}{\pi^2} = \frac{1-\pi}{\pi^2} \quad (12)$$

□

b)

By definition, in the sum of $X_1, X_2, \dots, X_n$, we will have exactly n successes, as each X_i is the number of trials up to and including the i -th success. The sum of such n random variables will require anything in the range $n, n+1, n+2, \dots$ trials to achieve n successes. This is exactly the model we are trying to describe with the negative binomial distribution. Let $Y_n = \sum_{i=1}^n X_i$, then we have by i.i.d, that

$$E[Y_n] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = nE[X_1] = \frac{n}{\pi} \quad (13)$$

Similarly for the variance

$$\text{Var}(Y_n) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = n\text{Var}(X_1) = \frac{n(1-\pi)}{\pi^2} \quad (14)$$

c)

Recall that the expected value of a product of independent random variables factorizes

$$X \perp Y \implies E[XY] = E[X]E[Y]$$

and if X is a random variable, then $f(X)$ is also a random variable, thus we have that $E[f(X)g(Y)] = E[f(X)]E[g(Y)]$ for any functions f and g .

We have that the moment generating function of Y_n is given by

$$M_{Y_n}(t) = E[e^{tY_n}] = E[e^{t\sum_{i=1}^n X_i}] = E\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n E[e^{tX_i}] = \prod_{i=1}^n M_{X_i}(t) = (M_{X_1}(t))^n. \quad (15)$$

Using the chain rule, we have that the derivative of $M_{Y_n}(t)$

$$\frac{d}{dt} M_{Y_n}(t) = \frac{d}{dt} (M_{X_1}(t))^n = n(M_{X_1}(t))^{n-1} \frac{d}{dt} M_{X_1}(t). \quad (16)$$

As such, we can find the cumulant generating function of Y_n by taking the logarithm of the moment generating function

$$K_{Y_n}(t) = \ln(M_{Y_n}(t)) = \ln((M_{X_1}(t))^n) = n \ln(M_{X_1}(t)). \quad (17)$$

We know the value of $M_{X_1}(t)$ from part a), thus

$$K_{Y_n}(t) = n \ln \left(\frac{\pi e^t}{1 - (1-\pi)e^t} \right) = n \left(t + \ln(\pi) - \ln(1 - (1-\pi)e^t) \right). \quad (18)$$

The first derivative of the cumulant generating function is given by

$$\frac{d}{dt} K_{Y_n}(t) = n \left(1 - \frac{(1-\pi)e^t}{1 - (1-\pi)e^t} \right), \quad (19)$$

evaluated at $t = 0$

$$\frac{d}{dt} K_{Y_n}(0) = n \left(1 - \frac{(1-\pi)}{1-(1-\pi)} \right) = n \left(1 + \frac{(1-\pi)}{\pi} \right) = \frac{n}{\pi}. \quad (20)$$

The variance of Y_n is given by the second derivative of the cumulant generating function, if we continue from (19), we have

$$\frac{d^2}{dt^2} K_{Y_n}(t) = n \frac{d}{dt} \left(\frac{(1-\pi)e^t}{1-(1-\pi)e^t} \right) = n \frac{(1-\pi)e^t}{((1-(1-\pi)e^t))^2}. \quad (21)$$

We can evaluate this at $t = 0$ to find the variance of Y_n

$$\frac{d^2}{dt^2} K_{Y_n}(0) = n \frac{(1-\pi)}{((1-(1-\pi))^2} = \frac{n(1-\pi)}{\pi^2} \quad (22)$$

□

d)

If we let $\pi = \frac{1}{2}$, then the probability of hitting $n+k$ trials before hitting n successes is given by the negative binomial distribution:

$$P(Y_n = n+k) = \binom{n+k-1}{k} \left(\frac{1}{2}\right)^n \left(1 - \frac{1}{2}\right)^k = \binom{n+k-1}{k} \left(\frac{1}{2}\right)^{n+k} = \binom{n+k-1}{k} \frac{1}{2^{n+k}} \quad (23)$$

Thus the sum

$$n \rightarrow \infty \quad \sum_{k=0}^n \binom{n+k-1}{k} \frac{1}{2^{n+k}} = n \rightarrow \infty \quad \sum_{k=0}^n P(Y_n = n+k)$$

This is a sum of disjoint events of independent random variables, thus we can express this as the probability of the union of these events, we note that we Y_n is in this union if and only if Y_n is in the range $[n, 2n]$, thus we have

$$\sum_{k=0}^n P(Y_n = n+k) = P\left(\bigcup_{k=0}^n Y_n = n+k\right) = P(\{Y_n = n\} \cup \{Y_n = n+1\} \cup \dots \cup \{Y_n = n+n\}) = P(n \leq Y_n \leq 2n) \quad (24)$$

Then from b), we know that we can express Y_n as a sum of n i.i.d. geometric random variables, as such

$$P(n \leq Y_n \leq 2n) = P(n \leq X_1 + X_2 + \dots + X_n \leq 2n) = P(1 \leq \bar{X} \leq 2). \quad (25)$$

We know from a) that X_i has finite mean and variance, thus we can apply the central limit theorem to find the limiting distribution of $\bar{X}$, let $\mu = \frac{1}{\pi} = 2$ and $\sigma^2 = \frac{1-\pi}{\pi^2}$, then

$$P(1 \leq \bar{X} \leq 2) = P\left(\frac{1-\mu}{\sigma/\sqrt{n}} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq \frac{2-\mu}{\sigma/\sqrt{n}}\right) = P\left(\frac{1-2}{\sigma/\sqrt{n}} \leq Z_n \leq \frac{2-2}{\sigma/\sqrt{n}}\right) = P\left(\frac{-1}{\sigma/\sqrt{n}} \leq Z_n \leq 0\right). \quad (26)$$

As we let $n \rightarrow \infty$, then

$$\lim_{n \rightarrow \infty} P\left(\frac{-1}{\sigma/\sqrt{n}} \leq Z_n \leq 0\right) = P(-\infty \leq Z \leq 0) = P(Z \leq 0) = \frac{1}{2} \quad (27)$$

□

e)

We now wish to find the exponential dispersion representation of Y_n for a fixed n . If we take the log of the PMF of Y_n and let $\beta = \binom{k-1}{n-1}$, we arrive at

$$\ln(P(Y_n = k)) = k \ln(1-\pi) + n (\ln(\pi) - \ln(1-\pi)) + \ln(\beta) \quad (28)$$

Now, we define $\theta = \ln(1-\pi)$, and $b(\theta) = -n (\ln(\pi) - \theta)$. It is usefule to note that $\theta = \ln(1-\pi) \implies e^\theta = 1-\pi \implies \pi = 1-e^\theta$. Thus we can rewrite $b(\theta)$ as

$$b(\theta) = -n (\ln(1-e^\theta) - \theta). \quad (29)$$

We then arrive at the following representation of the log PMF

$$\ln(P(Y_n = k)) = k\theta - b(\theta) + \ln(\beta). \quad (30)$$

This gives that the dispersion paratameter $\phi = 1$, and the canonical link function is given by

$$c(k, \phi) = \ln(\beta), \quad (31)$$

Thus, we see that the negative binomial distribution is a exponential dispersion model

$$P(Y_n = k) = \exp\left(\frac{k\theta - b(\theta)}{\phi} + c(k, \phi)\right). \quad (32)$$

We may find the mean by taking the derivative of $b(\theta)$

$$E[Y_n] = b'(\theta) = -n \left(\frac{-e^\theta}{1 - e^\theta} - 1 \right) = -n \left(\frac{-(1 - \pi)}{\pi} - 1 \right) = -n \left(\frac{-1}{\pi} + 1 - 1 \right) = \frac{n}{\pi}, \quad (33)$$

which is consistent with our previous results. We can find the variance by taking the second derivative of $b(\theta)$, since $\phi = 1$, we have

$$\text{Var}(Y_n) = b''(\theta) = -n \left(\frac{-e^\theta}{(1 - e^\theta)^2} \right) = -n \left(\frac{-(1 - \pi)}{\pi^2} \right) = \frac{n(1 - \pi)}{\pi^2}, \quad (34)$$

which is also consistent with our previous results. $\square$

f1)

We can use that the moment generating function of $\tilde{X}_\pi = \frac{\pi}{1-\pi}X$ is a linear transformation of the moment generating function of X , such that ¹

$$M_{\tilde{X}_\pi}(t) = M_X\left(\frac{\pi}{1-\pi}t\right). \quad (35)$$

As such,

$$M_{\tilde{X}_\pi}(t) = \frac{\pi e^{\frac{\pi}{1-\pi}t}}{1 - (1 - \pi)e^{\frac{\pi}{1-\pi}t}}. \quad (36)$$

Now let

$$g(\pi) = \pi e^{t \frac{\pi}{1-\pi}} \quad h(\pi) = 1 - (1 - \pi)e^{\frac{\pi}{1-\pi}t},$$

then we have

$$g'(\pi) = e^{t \frac{\pi}{1-\pi}} + \pi e^{t \frac{\pi}{1-\pi}} \frac{t}{(1 - \pi)^2} \quad h'(\pi) = e^{\frac{\pi}{1-\pi}t} - (1 - \pi)e^{\frac{\pi}{1-\pi}t} \cdot \frac{t}{(1 - \pi)^2}.$$

If we use l'Hopital's rule, we have that

$$\lim_{\pi \rightarrow 0} M_{\tilde{X}_\pi}(t) = \lim_{\pi \rightarrow 0} \frac{g'(\pi)}{h'(\pi)} = \frac{1}{1 - t} \quad (37)$$

Which the the moment generating function of an exponential distribution with rate parameter $\lambda = 1$. Thus we have that $\tilde{X}_\pi \rightarrow \text{Exp}(1)$ as $\pi \rightarrow 0$. If we then recall that we can express

$$\frac{\pi}{1 - \pi}Y_3 = \frac{\pi}{1 - \pi} \sum_{i=1}^3 X_i = \sum_{i=1}^3 \frac{\pi}{1 - \pi} X_i = \sum_{i=1}^3 \tilde{X}_{\pi i}, \quad (38)$$

and the fact that the sum of three independent exponential random variables with rate parameter $\lambda = 1$ is a gamma distribution with shape parameter $k = 3$ and rate parameter $\lambda = 1$, we have that for $\pi \rightarrow 0$, $\frac{\pi}{1 - \pi}Y_3 \rightarrow \text{Gamma}(3, 1)$. $\square$

¹https://en.wikipedia.org/wiki/Moment_generating_function